Adham Ehab Erfan Mahmoud

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Research Interests

Wireless Communications • 5G/6G Networks • Signal Processing & Estimation Theory • MIMO Systems • Beamforming & Codebook Design • Shannon Entropy • Mutual Information • Capacity Bounds • Machine Learning for Communications • Deep Learning-Based Channel Estimation • Predictive Beamforming • Autoencoders for End-to-End Communications

Profile

Communication and Electronics Engineering graduate specializing in the intersection of wireless networks and deep learning. With hands-on research experience in MIMO systems and neural codebook design, I am seeking a Master's scholarship to drive advanced, research-intensive projects in 5G/6G signal processing and communication theory.

Education

B.Sc. Communication and Electronics Engineering June 2026
Alexandria University — Faculty of Engineering *Alexandria, Egypt*

- GPA: **3.56 / 4.0** (*Excellent*)
- Graduation project (Excellent): Neural Codebook Design for MIMO Beam Management

General Secondary Education Certificate (Mathematics Stream) June 2021
Omar Ibn ElHatab *Alexandria, Egypt*

- Score: **90%**
- Core Focus: Advanced Mathematics, Mechanics, and Physics

Research Experience

Graduation Project — NBL for MIMO Beam Management 2025 – 2026
Alexandria University *Alexandria, Egypt*

- Developed a deep learning framework for SSB and CSI-RS codebook generation in multi-cell 5G networks to maximize spectral efficiency under strict feedback overhead limits.
- Simulated ray-traced multi-cell environments with overlapping base-station coverage using the **Sionna** library.
- Integrated initial access, beam refinement, and channel state information (CSI) feedback into a unified end-to-end learning pipeline.
- Evaluated key trade-offs in power allocation, interference management, and beam tracking accuracy across complex multi-cell layouts.
- *Manuscript in preparation*: Research paper based on project results targeted for submission.

R&D Volunteer 2024 – Present
University Center for Career Development (UCCD) *Alexandria, Egypt*

- Contributed to technical analysis, research documentation, and engineering-oriented reports.

Selected Projects

Neural Network-Based Jammer Frequency Band Optimization — *Physical Layer Security / Deep Learning*

- Designed a Deep Q-Network (DQN) agent to automatically target and jam frequency bands without knowing the communication hopping pattern beforehand.
- Simulated a Frequency Hopping Spread Spectrum (FHSS) link to test the jammer against various patterns, including pseudo-random and linear sweeps.
- Implemented experience replay, target networks, and an ϵ-greedy policy to train the model and balance frequency exploration.

Reconfigurable Intelligent Surfaces for 5G/6G Systems — *Wireless Communications / Research Survey*

- Surveyed RIS phase-shift design, analyzing trade-offs between alternating optimization (AO) and deep learning.

Industry Experience

AI Model Developer 2024 – 2025
Outlier / Aligner *Remote*

- Developed and evaluated AI model responses in mathematics, algorithm design.

Embedded Systems Intern Summer 2024
Siemens Egypt *Egypt*

- Worked with AVR/ARM microcontroller architectures, startup code, and communication protocols (UART, SPI).
- Programmed ESP32 for automated data acquisition and wireless cloud streaming.

Technical Skills

Expert	MATLAB, Digital Communications, MIMO Systems, Beamforming
Advanced	Python, Signal Processing, Machine Learning (NumPy, scikit-learn, pytorch, tensorflow)
Intermediate	C, C++, Embedded Systems (AVR, ARM), UART/SPI, IoT Development
Tools	LaTeX, ROS, PCB Design, Git
Languages	Arabic (Native), English (Professional — IELTS Academic 6.0)

Certifications

- **IELTS Academic** — Band 6.0
- **Deep Learning** — NVIDIA
- **Machine Learning** — Generative Technology Center
- **Embedded Systems Essentials** — SIEMENS